Solution:

Calculation of price index number by simple average of price relatives:

Item	Price in 2001 (₹) p_0	Price in 2002 (₹) p_1	$p = \frac{p_1}{p_0} \times 100$	log p
A	6	10	166.7	2.2219
B	2	2	100.0	2.0000
C	4	6	150.0	2.1761
D	10	12	120.0	2.0792
E	8	12	150.0	2.1761
			$\sum p = 686.7$	$\sum \log p = 10.6533$

(i) Price relative index number based on arithmetic mean:

$$P_{01} = \frac{\sum \frac{p_1}{p_0} \times 100}{N} = \frac{\sum p}{N} = \frac{686.7}{5} = 137.34$$

(ii) Price relative index number based on geometric mean:

$$P_{01} = \text{antilog}\left(\frac{\sum \log p}{N}\right) = \text{antilog}\left(\frac{10.6533}{5}\right)$$
$$= \text{antilog}\,(2.13066)$$
$$= 135.1$$

Hence, the price index number based on arithmetic mean and geometric mean for the year 2002 are 137.34 and 135.1 respectively.

Example 6.4

Construct simple average price relative index number using arithmetic mean for the year 2012 for the following data showing the profit from various categories sold out in departmental stores.

Profit (per week)	2010	2012
Groceries	150600	170800
Cosmetics	70000	82000
Stationery items	12000	10800
Utensils	20000	18600

Solution: Index number uning Arithmertic Mean of price relatives

	Profit in 2010 (p_0)	Profit in 2012 (p_1)	p_1/p_0 x 100
Groceries	150600	170800	$\frac{170800}{150600} \times 100 = 11341$
Cosmetics	70000	82000	$\frac{82000}{70000} \times 100 = 117.14$
Stationery items	12000	10800	$\frac{10800}{12000} \times 100 = 90.00$
Utensils	20000	18600	$\frac{18600}{20000} \times 100 = 93.00$
		Total	413.55

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